I-MAP: LOGBOOK (AI USAGE)

TSMK

Universiteit van Antwerpen
Toegepaste Ingenieurswetenschappen (Elektronica-ICT)

LLM request

⚠ Claude was used for this prompt.

i'm using a NEO 6M gps module. It says online that it only works with baudrate 9600, could i possibly increase it?

LLM resopnse

Yes, you can increase the baud rate of the NEO-6M beyond 9600! Here's how it works:

How to Change the Baud Rate

The NEO-6M's default baud rate is 9600, but it supports higher rates. You change it by sending a **UBX configuration command** over the serial interface.

Supported baud rates: 4800, 9600, 19200, 38400, **57600**, **115200**, 230400

Method 1: UBX CFG-PRT Command (Recommended)

Send this byte sequence at 9600 to switch to **115200**:

```
0xB5 0x62 0x06 0x00 0x14 0x00  ← Header + Class + ID + Length
0x01                    ← Port ID (UART1)
0x00                    ← Reserved
0x00 0x00                 ← TX ready (disabled)
0xD0 0x08 0x00 0x00           ← UART mode (8N1)
0x00 0xC2 0x01 0x00           ← Baud rate = 115200 in little-endian
0x07 0x00                 ← Input protocols (UBX+NMEA+RTCM)
0x03 0x00                 ← Output protocols (UBX+NMEA)
0x00 0x00                 ← Flags
0x00 0x00                 ← Reserved
0xBC 0x5E                  ← Checksum (CK_A, CK_B)
```

After sending this, **immediately switch your host's baud rate** to 115200 — the module changes instantly.

Method 2: u-center Software

If you have a USB-to-TTL adapter, use **u-blox's free u-center** tool:

1. Connect at 9600
2. Go to **View → Configuration View → PRT (Ports)**
3. Change the baud rate and click **Send**
4. Optionally click **CFG → Save config** to persist it

LLM request

could i do this via linux, since i only have a debain 13 system?

LLM resopnse

Yes, Linux works great for this! You can send the UBX command directly via the terminal.

Step 1 — Find the Arduino's Port

bash

```
ls /dev/ttyUSB* /dev/ttyACM*
```

It'll likely show up as `/dev/ttyACM0`.

Step 2 — Upload the Empty Sketch

You need the Arduino IDE or `arduino-cli` to upload the passthrough sketch. The quickest way on Debian:

bash

```
sudo apt install arduino-cli
arduino-cli core install arduino:avr

# Save the empty sketch
mkdir -p ~/gps_config && cat > ~/gps_config/empty/empty.ino << 'EOF'
void setup() {}
void loop() {}
EOF

# Upload it
arduino-cli compile --fqbn arduino:avr:uno ~/gps_config/empty
arduino-cli upload -p /dev/ttyACM0 --fqbn arduino:avr:uno ~/gps_config/empty
```

Step 3 — Send the UBX Command via Python

Once the empty sketch is uploaded and the NEO-6M is wired to pins 0/1:

bash

```
sudo apt install python3-serial
```

Then create and run this script:

python

```
# ~/gps_config/set_baud.py
import serial
import time

PORT = "/dev/ttyACM0"

# UBX CFG-PRT: switch UART1 to 115200
cfg_prt = bytes([
    0xB5, 0x62, 0x06, 0x00, 0x14, 0x00,
    0x01, 0x00, 0x00, 0x00,
    0xD0, 0x08, 0x00, 0x00,
    0x00, 0xC2, 0x01, 0x00,  # 115200
    0x07, 0x00, 0x03, 0x00,
    0x00, 0x00, 0x00, 0x00,
```

```
    0xBC, 0x5E
])

# UBX CFG-CFG: save to flash
cfg_save = bytes([
    0xB5, 0x62, 0x06, 0x09, 0x0D, 0x00,
    0x00, 0x00, 0x00, 0x00,
    0xFF, 0xFF, 0x00, 0x00,
    0x00, 0x00, 0x00, 0x00,
    0x17, 0x31, 0xBF
])

print("Connecting at 9600...")
with serial.Serial(PORT, 9600, timeout=1) as ser:
    time.sleep(2)  # let port settle

    print("Sending CFG-PRT (switch to 115200)...")
    ser.write(cfg_prt)
    time.sleep(0.1)

print("Reopening at 115200...")
with serial.Serial(PORT, 115200, timeout=1) as ser:
    time.sleep(0.5)

    print("Sending CFG-CFG (save to flash)...")
    ser.write(cfg_save)
    time.sleep(0.2)

    # Verify it's working by reading NMEA output
    print("\nReading GPS output (Ctrl+C to stop):")
    while True:
        line = ser.readline()
        if line:
            print(line.decode('ascii', errors='replace').strip())
```

bash

```
python3 ~/gps_config/set_baud.py
```

If you see NMEA sentences like `$GPRMC`, . . . scrolling by at the end, it worked!

LLM request

What byte arrays do i need to send to increase the baud rate and refresh speed? Also are there data points that i can remove?

LLM resopnse

```
        /** @brief UBX CFG-PRT command: Switch baud rate to 115200.
*/
        const byte UBX_BAUD[28] = {
            0xB5, 0x62, 0x06, 0x00, 0x14, 0x00,
            0x01, 0x00, 0x00, 0x00,
            0xD0, 0x08, 0x00, 0x00,
            0x00, 0xC2, 0x01, 0x00,  // 115200 baud
            0x07, 0x00, 0x03, 0x00,
            0x00, 0x00, 0x00, 0x00,
            0xC0, 0x7E
        };

        /** @brief UBX CFG-RATE command: Set update rate to 10Hz.
*/
        const byte UBX_10HZ[14] =
{0xB5,0x62,0x06,0x08,0x06,0x00,0x64,0x00,0x01,0x00,0x01,0x00,0x7A,0
x12};

        /** @brief UBX CFG-MSG command: Disable GLL sentence. */
        const byte DIS_GLL[16] =
{0xB5,0x62,0x06,0x01,0x08,0x00,0xF0,0x01,0x00,0x00,0x00,0x00,0x00,0
x00,0x00,0x2A};
        /** @brief UBX CFG-MSG command: Disable GSA sentence. */
        const byte DIS_GSA[16] =
{0xB5,0x62,0x06,0x01,0x08,0x00,0xF0,0x02,0x00,0x00,0x00,0x00,0x00,0
x00,0x01,0x31};
        /** @brief UBX CFG-MSG command: Disable GSV sentence. */
        const byte DIS_GSV[16] =
{0xB5,0x62,0x06,0x01,0x08,0x00,0xF0,0x03,0x00,0x00,0x00,0x00,0x00,0
x00,0x02,0x38};
        /** @brief UBX CFG-MSG command: Disable VTG sentence. */
        const byte DIS_VTG[16] =
{0xB5,0x62,0x06,0x01,0x08,0x00,0xF0,0x05,0x00,0x00,0x00,0x00,0x00,0
x00,0x04,0x46};
```